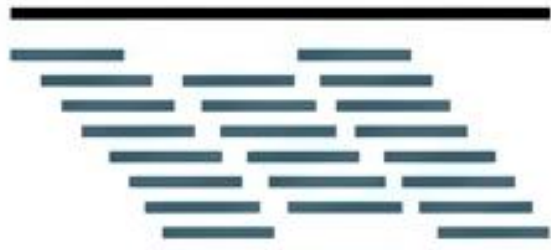


Genomic Reconstruction & Variant Extraction

Pipeline Goal: Reconstruct the evolutionary relationships of bacterial populations by mapping sequence data to a reference, extracting pure vertical descent signals, and clustering isolates for high-resolution genomic surveillance.



Step 1: Reference Mapping & Variant Calling

Objective: Map raw sequence reads against a reference genome to identify variants and build a consensus pseudosequence.

Importance: Forms the foundation of comparative genomics by precisely cataloging the genetic differences (SNPs/INDELs) between the isolate and the baseline reference.

Tool: Snippy (integrating BWA & FreeBayes)

Input: FASTQ paired-end reads & Reference Genome (FASTA)

Output: BAM (alignments), VCF (variants), and a core full genome alignment



Step 2: Alignment Masking & Cleaning

Objective: Mask uncertain or low-confidence genomic regions across the alignment with 'N's.

Importance: Prevents areas of low sequence coverage from generating false positive/negative variant calls, ensuring robust downstream evolutionary analysis.

Tool: snippy-clean_full_aln

Input: Core full genome alignment

Output: Cleaned whole-genome alignment



Step 3: Variable Site (SNP) Extraction

Objective: Strip away identical, invariant DNA sequences to isolate only the polymorphic sites.

Importance: Drastically reduces computational overhead for complex phylogenetic algorithms without sacrificing the core evolutionary signal.

Tool: snp-sites

Input: Cleaned whole-genome alignment

Output: SNP-only multiple sequence alignment



Continued on Next Slide

Snippy outputs (*.txt)

Reference *V. cholerae*_O1_biovar_ElTor_str_N16961
ReferenceSize 4033501
Software: snippy 4.6.0

	New sample-1	New sample 2	CTMA_1441
Variant-COMPLEX	29		2
Variant-DEL	10		8
Variant-INS	11		3
Variant-MNP	3	2	
Variant-SNP	204	87	143
VariantTotal	257	87	156

Snippy outputs (*.tab)

CHROM	POS	TYPE	REF	ALT	EVIDENCE	FTYPE	STRAND
PRODUCT							
AE003852		27804	snp	C	T	T:54 C:0	
AE003852		34254	snp	C	T	T:50 C:0	
AE003852		167268	snp	G	A	A:38 G:1	
AE003852		185099	snp	G	T	T:45 G:0	
AE003852		185958	del	CT	C	C:36 CT:0	
AE003852		254869	snp	T	C	C:44 T:0	
AE003852		281442	snp	C	T	T:35 C:0	
AE003852		286337	snp	C	T	T:40 C:0	
AE003852		347722	snp	T	C	C:39 T:0	
AE003852		364550	snp	C	A	A:37 C:0	
AE003852		376379	snp	C	³ T	T:48 C:0	
AE003852		451435	snp	A	T	T:38 A:0	
AE003852		467111	del	CGACCCTA	C	C:15 CGACCCTA:0	

Looking at variants in detail with `snippy-vcf_report`

If you run Snippy with the `--report` option it will automatically run `snippy-vcf_report` and generate a `snps.report.txt` which has a section like this for each SNP in `snps.vcf`:

```
~~~~~
>LBB_contig000001:10332 snp A=>T DP=7 Q=66.3052 [7]

          10301      10311      10321      10331      10341      10351      10361
tcttctccgagaagggaatataatttaaaaaaattcttaaataattcccttccctcccggtataaaaaattcttcgcttat
.....T.....
//////////t//////////t,t//////////g//////////g//////////t,
//////////T.....A.....A.....
.....A.....A.....T.....C.....
.....A.....C..C.....CT.....TA.....
,a,,,a,,,,,t,t,,g,,,,,
,,,ga,,,c,,,t,,,g,,,t,,,,,
.....T.C.....G.....G.....
,,,,,a,,,,,a,,,,,
```

BAM visualization

File

Entries

Select

View

Goto

Edit

Create

Run

Graph

Display

Entry:

☒ Vibrio_cholerae_01_biovar_eltor_str_N16961_v2.fa

Nothing selected

<<

>>

AE003852

800

1600

2400

3200

4000

4800

5600

6400

7200

8000

8800

9600

10400

11200

12000

<<

>>

R V I K Y I # R S I + R S F Y + I Y Y # G A G S L W I S E K * S T R S C D S E G S D R V I N H * S V Q G L A G I K N L C Y T Q P P W

G S L N I Y K D L Y R D L F I R S T I K E Q D L C G # V K N D Q Q D H A I Q K D Q I V * S T T D L F K D + L G S K T Y V I H S H L G

. G H # I Y I K I Y I E I F L L D L L L R S R I F V D K * K M I N K I M R F R R I R S C D Q P L I C S R I S W D Q K P M L Y T A T L

AGGGTCATTAATATATATAAAGATCTATATAGAGATCTTTTATTAGATCTACTATTAAGGAGCAGGATCTTTGTGGATAAGTGAAAAATGATCAACAAGATCATGCGATTGAGAAGGATCAGATCGTGTGATCAACCACTGATCTGTTCAAGGATTAGCTGGGATCAAAAACCTATGTTATACACAGCCACCTTGG

20

40

60

80

100

120

140

160

180

TCCAGTAATTTATATATATTTCTAGATATATCTCTAGAAAAATAATCTAGATGATAATTCCTCGTCCTAGAAACACCTATTCACTTTTACTAGTTGTTCTAGTACGCTAAGTCTTCTAGTCTAGCACACTAGTTGGTGACTAGACAAGTTCCTAATCGACCTAGTTTTTGGATACAATATGTGTCGGTGGAACC

. T M L Y I Y L D I Y L D K # # I + + # P A P D K H I L S F H D V L D H S E S P D S R T I L W Q D T * P N A P I L F R H # V C G G Q

P D N F I Y L S R Y L S R K I L D V I L S C S R Q P Y T F F S * C S * 5 A I * F S * I T H D V V S R N L S # S P D F V + T I C L W R P

P * # I Y I F I + I S I K K N S R S N L L L I K T S L H F I I L L I M R N L L I L D H S * G S I Q E L I L Q S * F G I N Y V A V K P

<<

>>

fasta_record

1 2961182

AE003852 Vibrio cholerae 01 biovar eltor str. N16961 chromosome I, complete sequence.

fasta_record

2961183 4033501

AE003853 Vibrio cholerae 01 biovar eltor str. N16961 chromosome II, complete sequence.

A...

-

□

×

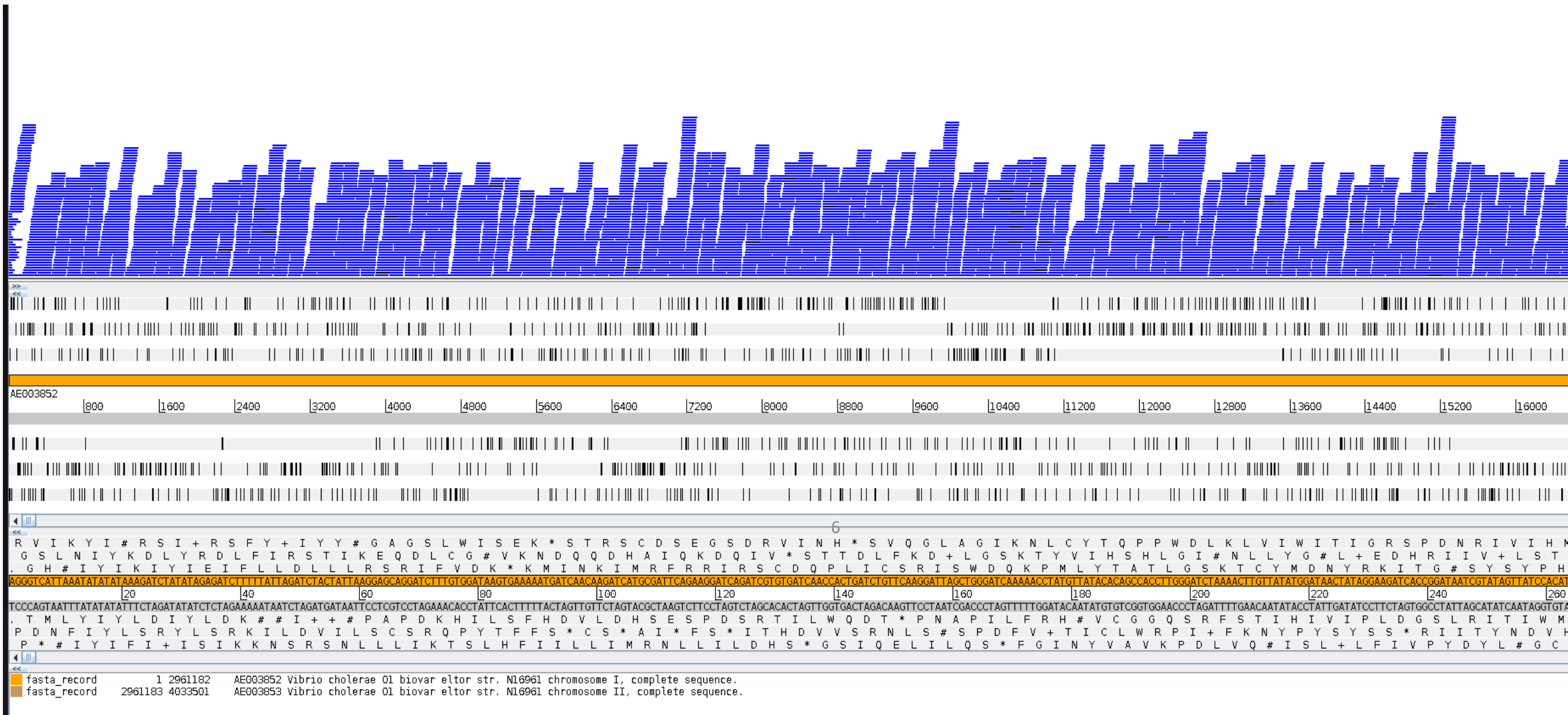
BAM / CRAM / VCF file:

Select...

Add More

OK

BAM visualization



BAM visualization

ТСД

TCAAGGATTAGCTGG

[illegible][illegible]

Figure 6

Figure 6

Figure 1. The effect of the *in vitro* and *in vivo* tests on the α -glucosidase activity of the *in vitro* and *in vivo* tests.

Q U E R Y O U I R N E

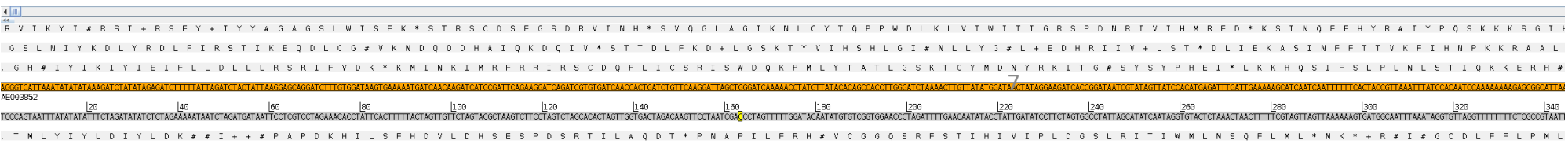
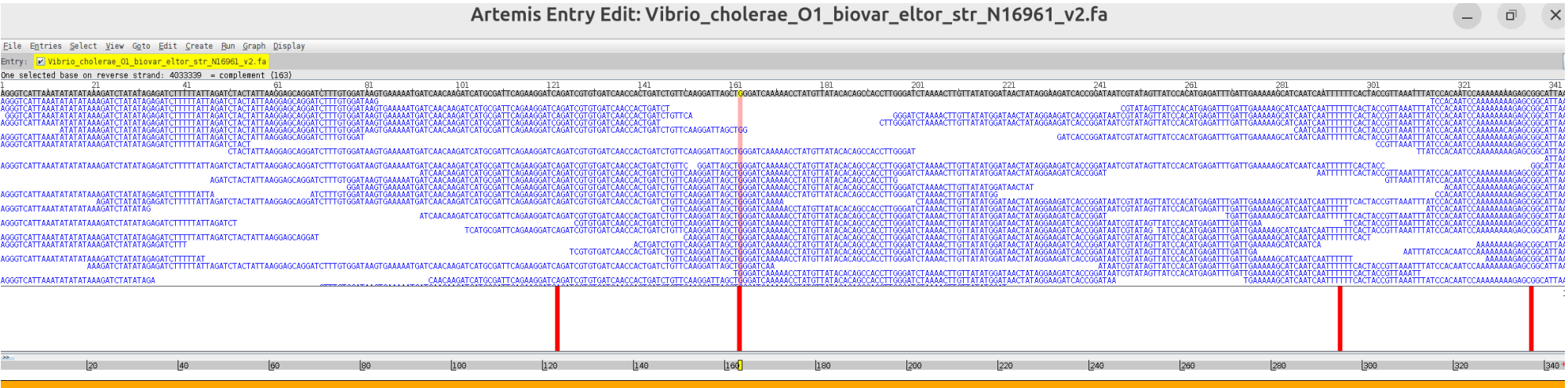
F K D + L G S K T Y

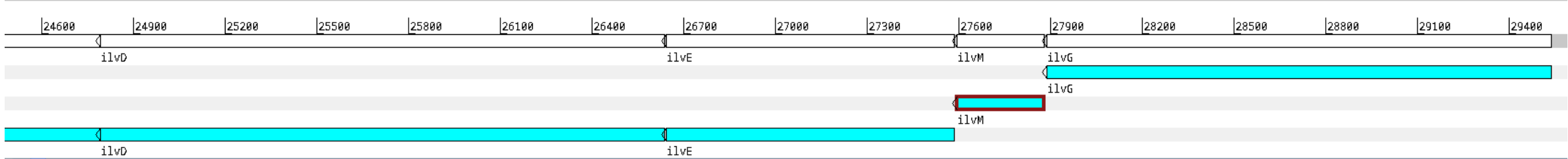
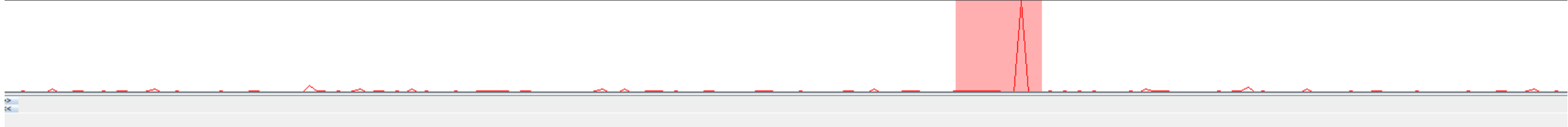
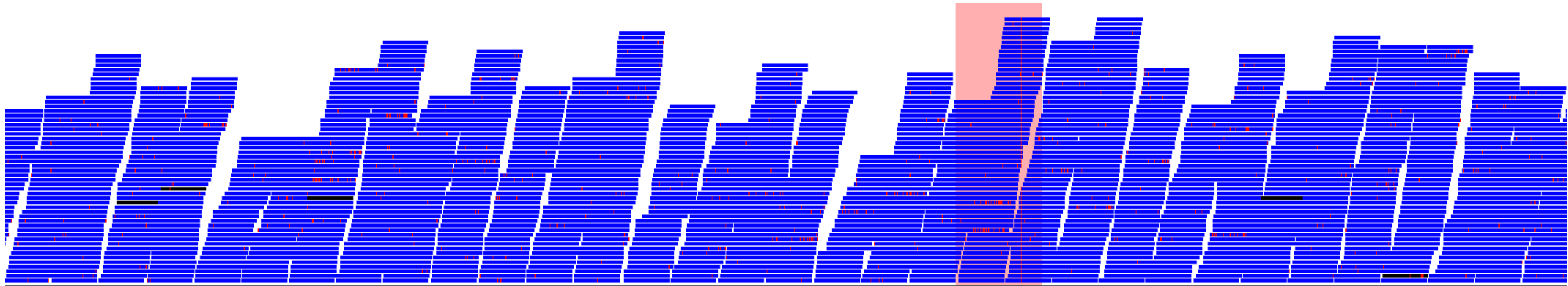
A B C D E F G H I

T	C	A	G	G	A	T	T	A	G	C	T	G	G	G	A	T	C	A	A	A	A	C	C	T	A
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

160

AGTTCTTAATCGACCTAGTTTTTGGAT





R V I K Y I # R S I + R S F Y + I Y Y # G A G S L W I S E K * S T R S C D S E G S D R V I N H * S V Q G L A G I K N L C Y T Q P P W D L K L V I W I T I G R S P D I
G S L N I Y K D L Y R D L F I R S T I K E Q D L C G # V K N D Q Q D H A I Q K D Q I V * S T T D L F K D + L G S K T Y V I H S H L G I # N L L Y G # L + E D H R I
G H # I Y I K I Y I E I F L L D L L L R S R I F V D K * K M I N K I M R F R R I R S C 8 D Q P L I C S R I S W D Q K P M L Y T A T L G S K T C Y M D N Y R K I T G #
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P D N F I Y L S R Y L S R K I L D V I L S C S R Q P Y T F F S * C S * A I * F S * I T H D V V S R N L S # S P D F V + T I C L W R P I + F K N Y P Y S Y S S * R I :
P * # I Y I F I + I S I K K N S R S N L L L I K T S L H F I I L L I M R N L L I L D H S * G S I Q E L I L Q S * F G I N Y V A V K P D L V Q # I S L + L F I V P Y
T M L Y I Y L D I Y L D K # # I + + # P A P D K H I L S F H D V L D H S E S P D S R T I L W Q D T * P N A P I L F R H # V C G G Q S R F S T I H I V I P L D G S L

CDS	22028	23008	c	Alcohol dehydrogenase
gene	23256	24788	c	
CDS	23256	24788	c	threonine ammonia-lyase, biosynthetic
gene	24792	26633	c	
CDS	24792	26633	c	dihydroxy-acid dehydratase
gene	26643	27581	c	
CDS	26643	27581	c	branched-chain-amino-acid transaminase
gene	27593	27877	c	
CDS	27593	27877	c	acetolactate synthase 2 small subunit

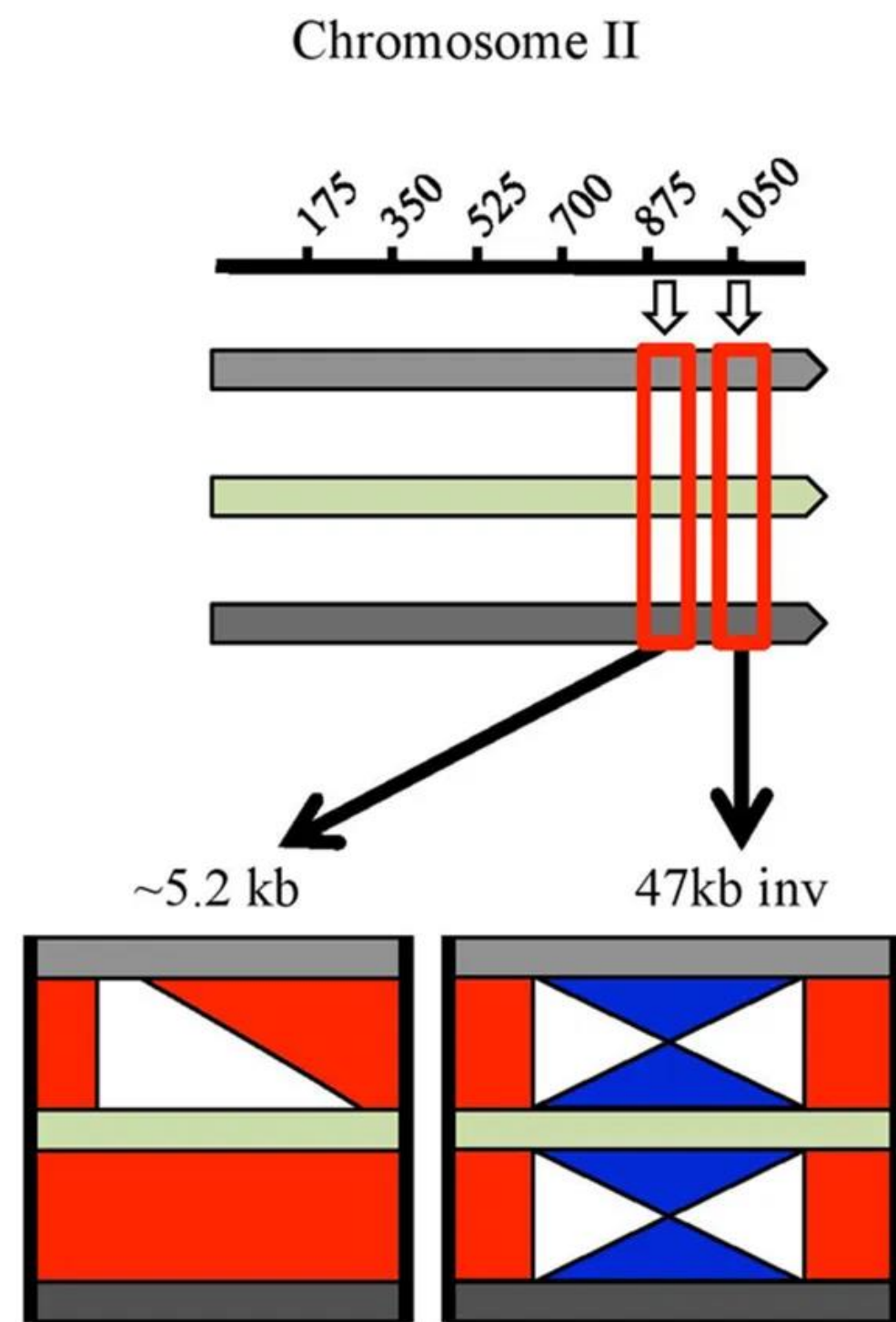
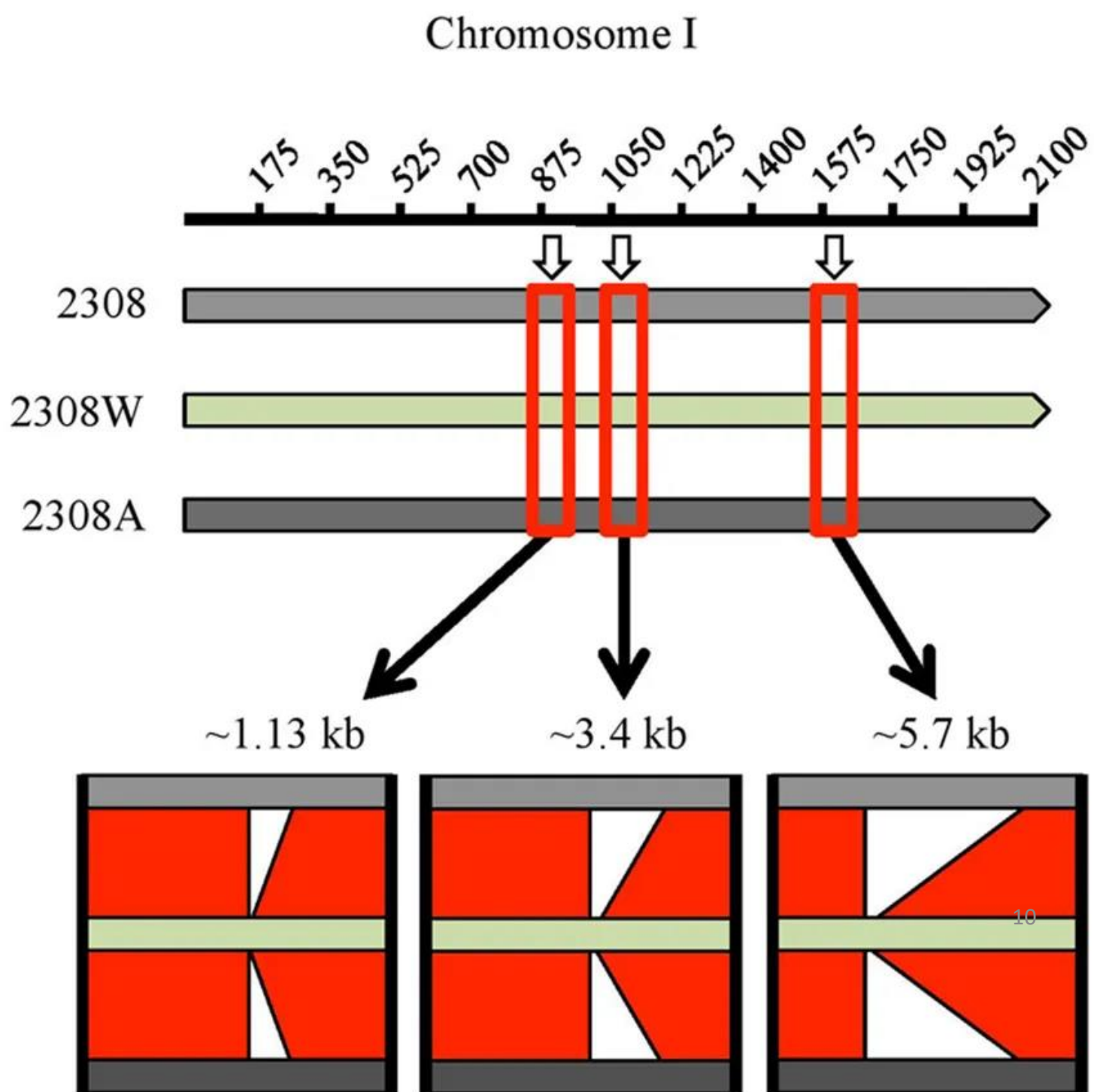
Core alignments

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TGGATGCTCGG
```

```
>CTMA_1441_mapped
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GGTGTTACTTGGAGCATTAGGGGCACCCCGCAAACCTCTACTATTGTCAATCACACCGGCGACTAGAGAAAGGACAGTCCC
GGGAAATTATTATTATTAACGTATGGACCAAAAAAGACAAGTCAATATCCAGCCCGACTTCAGCTGTCCTA
```

1991 nt

Nucleotide positions shared by all strains, with variants
No indels because they are not present in the reference



Evolutionary Inference & Contextualization

Pipeline Goal: Translate refined genomic variants into robust statistical models of ancestral descent, automatically correcting for horizontal gene transfer, to produce actionable epidemiological dashboards.



Step 4: Maximum Likelihood Phylogeny

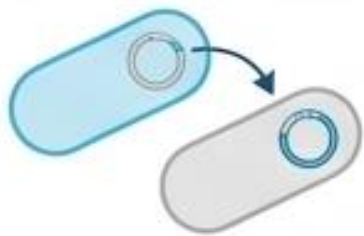
Objective: Calculate a maximum-likelihood phylogenetic tree to reconstruct the ancestral relationships among the samples.

Importance: Translates raw genetic differences into a statistical model of evolutionary history and genetic distance.

Tool: IQ-TREE2

Input: SNP-only alignment

Output: Unrooted maximum-likelihood phylogenetic tree (Newick format)



Step 5: Recombination Accounting

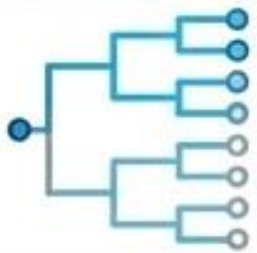
Objective: Identify and mask genomic regions introduced via horizontal gene transfer rather than vertical descent.

Importance: Recombination events introduce massive amounts of false SNPs that artificially inflate genetic distances and distort true clonal relationships.

Tool: Gubbins

Input: Cleaned full alignment

Output: Recombination-masked SNP FASTA & adjusted phylogeny



Step 6: Tree Rooting

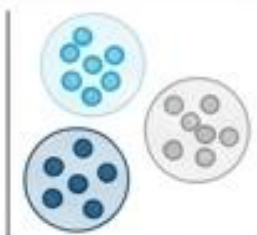
Objective: Root the phylogeny at the exact midpoint of the longest path between any two tips.

Importance: Applies essential evolutionary directionality to the data, implying a common ancestor and establishing a timeline of divergence.

Tool: Custom Script (midpoint.root.py)

Input: Recombination-adjusted unrooted tree

Output: Midpoint-rooted phylogenetic tree



Step 7: Population Clustering

Objective: Statistically group genomes into distinct evolutionary sub-populations based on genetic similarity.

Importance: Translates complex branch lengths into discrete, actionable epidemiological groupings vital for outbreak investigation.

Tool: fastBAPS

Input: Recombination-filtered SNP alignment & phylogenetic tree

Output: Cluster assignments (CSV metadata)



Step 8: Interactive Visualization

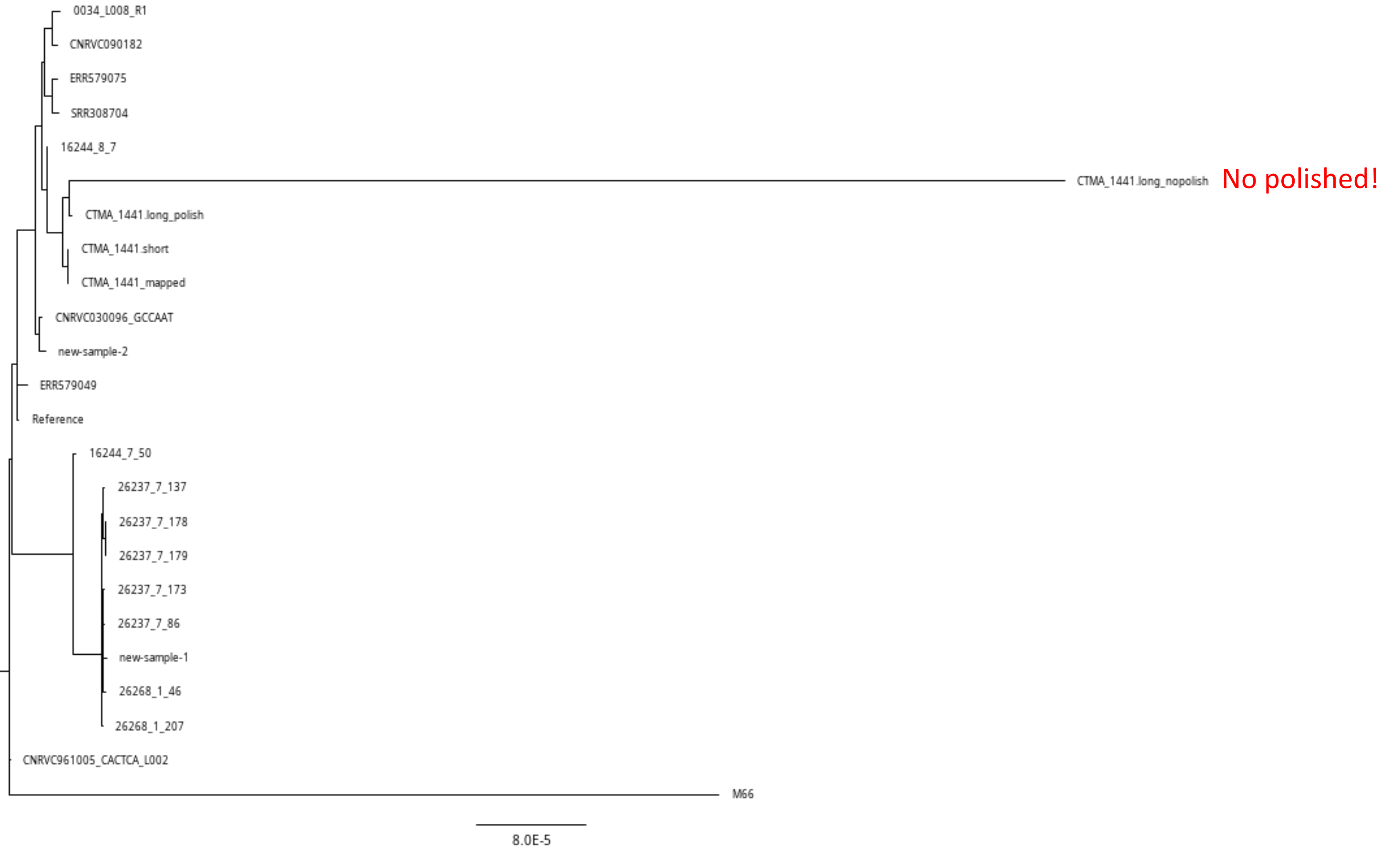
Objective: Visually integrate the phylogeny with recombination blocks and epidemiological metadata.

Importance: Allows scientists to seamlessly cross-reference evolutionary lineages with geographic, temporal, and clinical data to draw final conclusions.

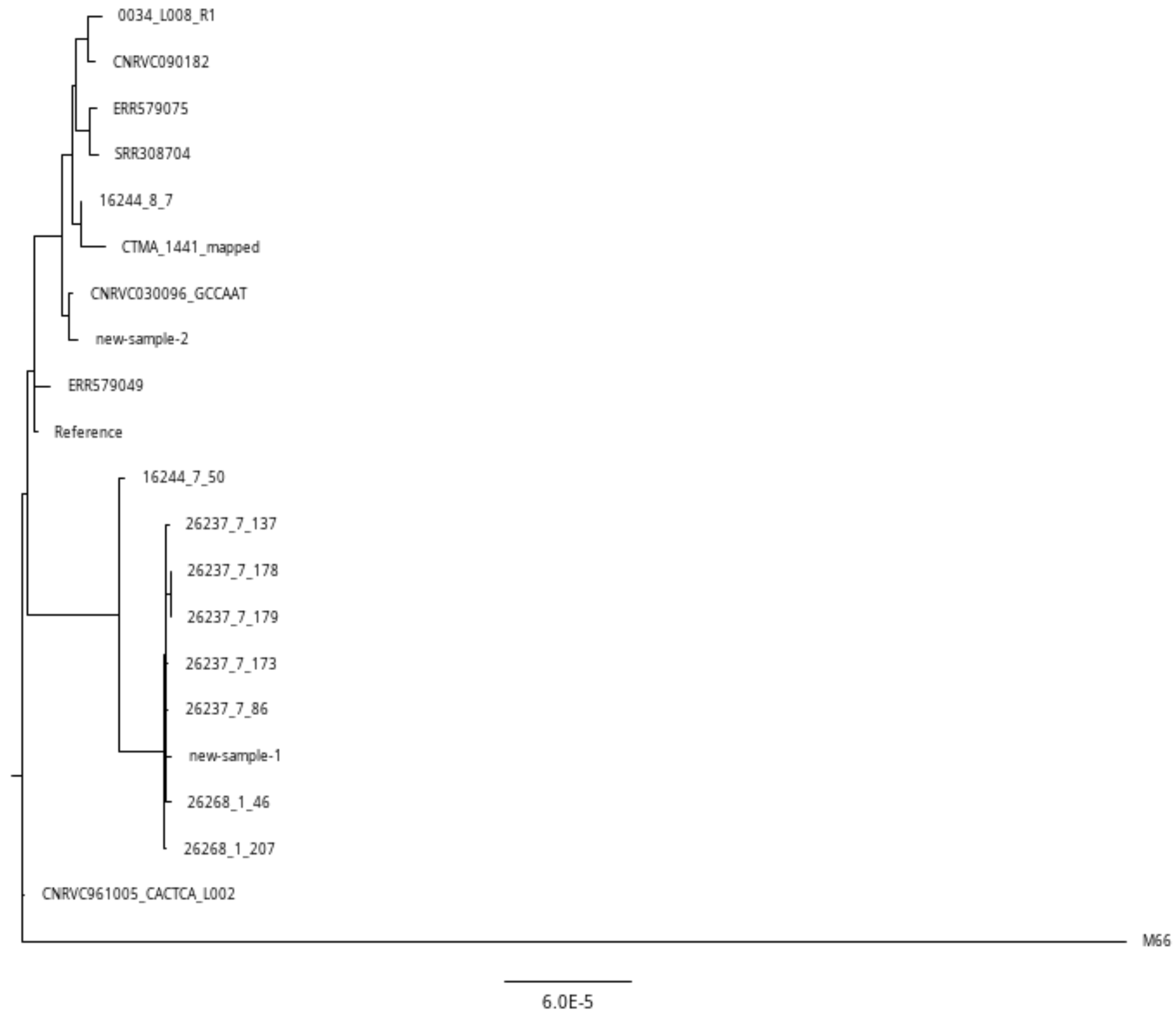
Tools: Microreact, Phandango, or FigTree

Input: Rooted tree, Cluster CSV, Recombination GFF

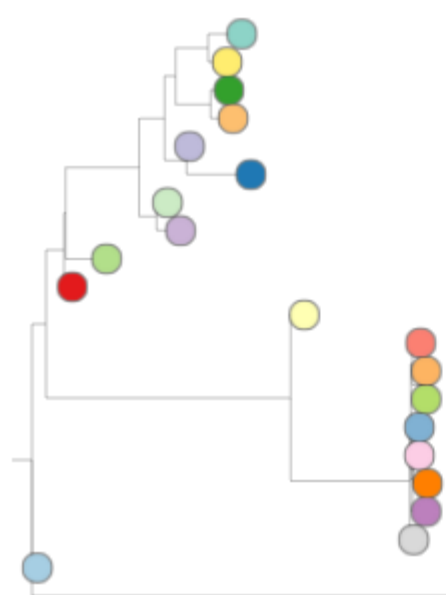
Output: Interactive, multi-layered visual dashboards



After removing poor quality genome



Tree 🗄



0.000032

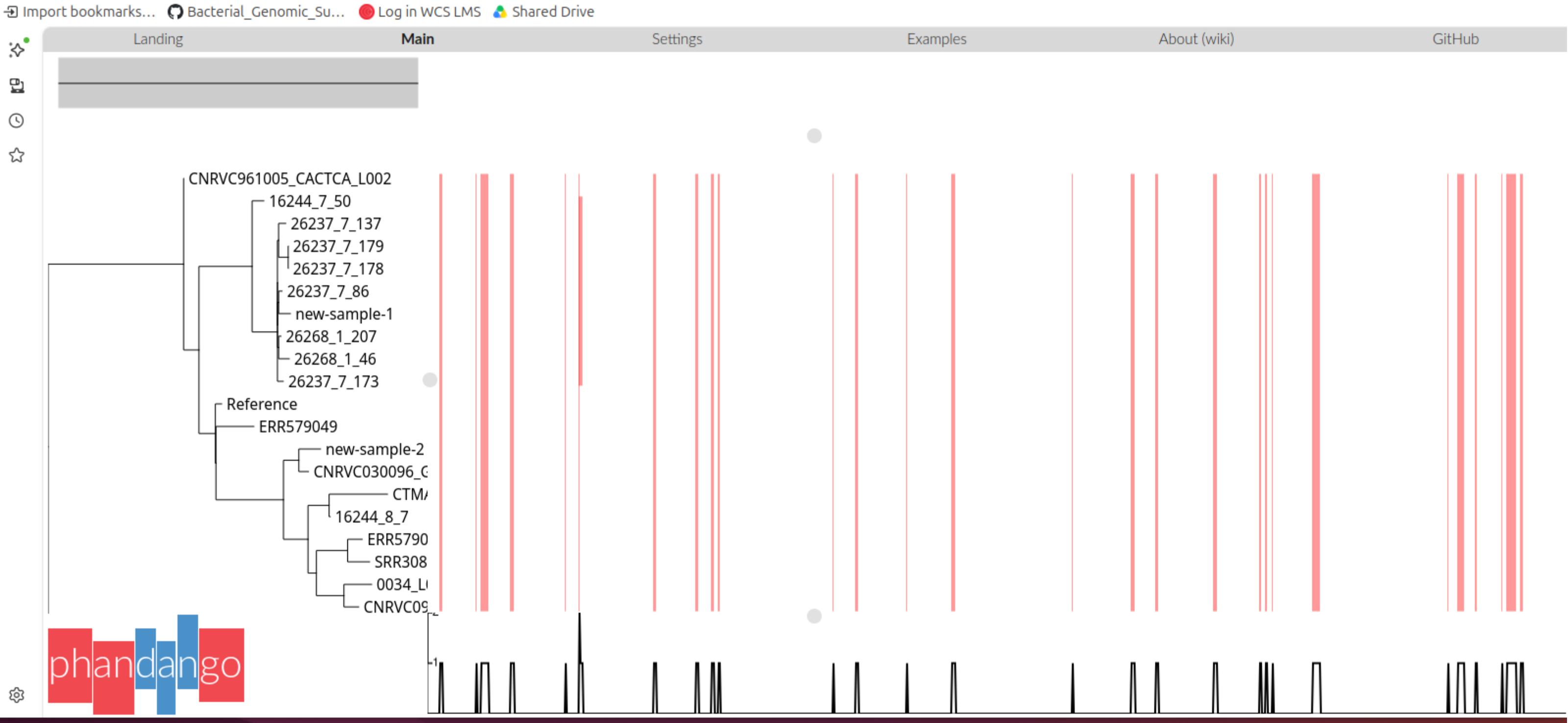
PhyloCanvas.js

Metadata 🗄

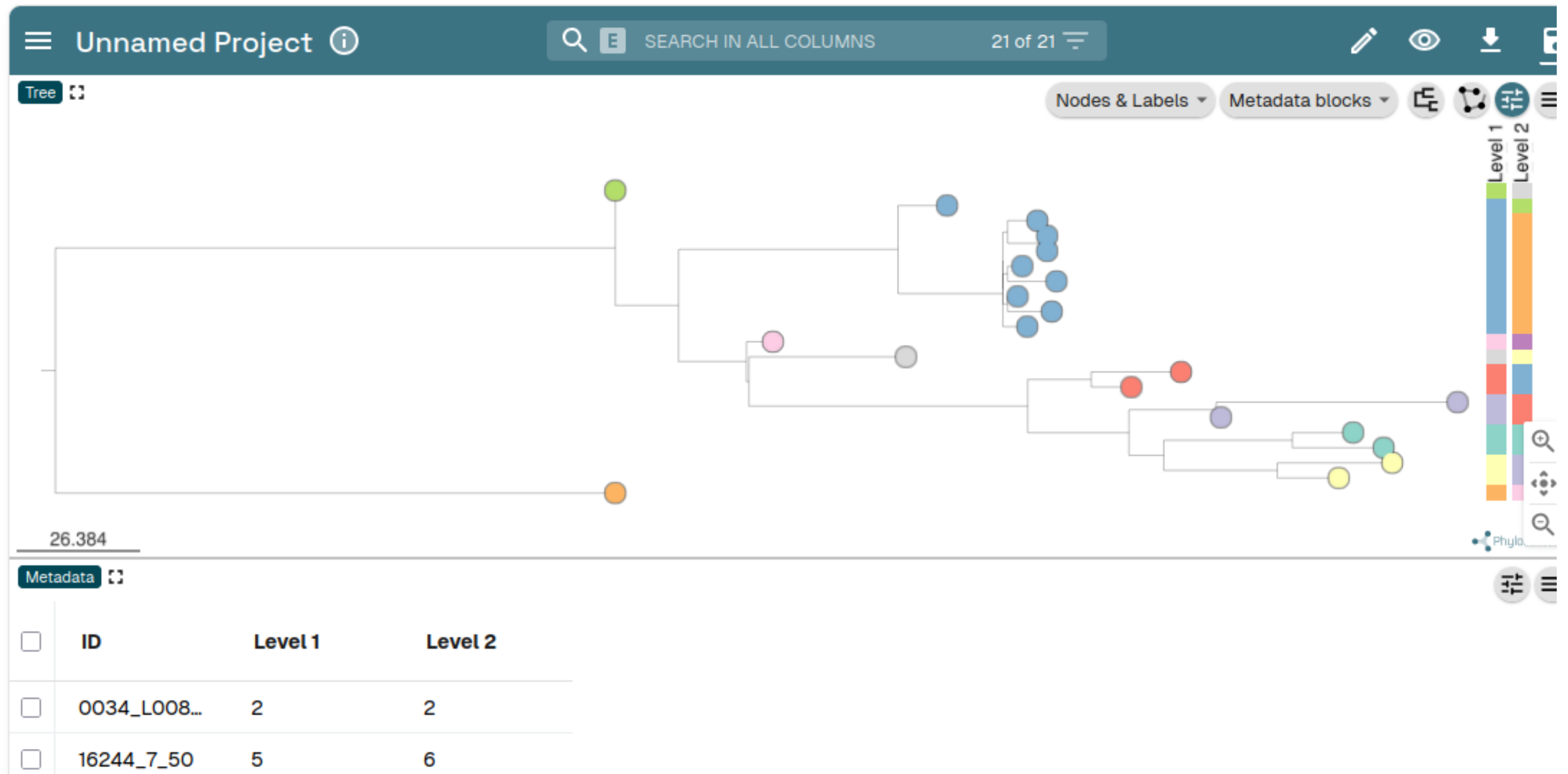


☐	Id
☐	CNRVC9610...
☐	M66
☐	Reference
☐	16244_7_50

After removing recombination



fastBAPS



1

2

3

